

Morphological Image Processing

Opening , Closing & HMT

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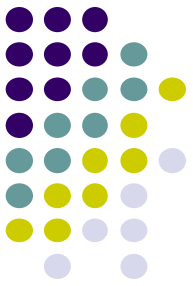
Faculty of Computer Engineering

University of Isfahan



Remind Erosion and Dilation

Usefulness:



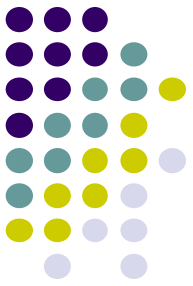
- **Erosion**

- removal of structures of certain shape and size, given by SE

- **Dilation**

- filling of holes of certain shape and size, given by SE

Dilation expands and Erosion shrinks.



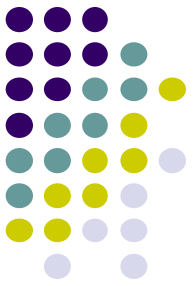
Erosion Then Dilation : eliminating irrelevant detail



a b c

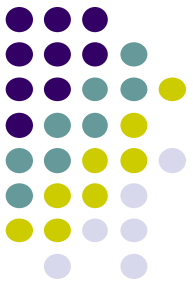
FIGURE 9.7 (a) Image of squares of size 1, 3, 5, 7, 9, and 15 pixels on the side. (b) Erosion of (a) with a square structuring element of 1's, 13 pixels on the side. (c) Dilation of (b) with the same structuring element.

structuring element $B = 13 \times 13$ pixels of gray level 1



Combining erosion and dilation

- WANTED:
 - remove structures / fill holes
 - without affecting remaining parts
- SOLUTION:
 - combine erosion and dilation (using same SE)



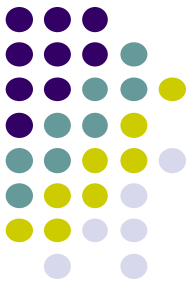
Opening

The opening of set A by structuring element B , denoted $A \circ B$, is defined as

$$A \circ B = \bigcup \left\{ (B)_z \mid (B)_z \subseteq A \right\}$$

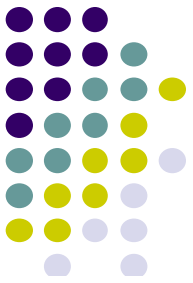
It is equivalent to Erosion followed by Dilation, denoted $\circ$

$$A \circ B = (A \ominus B) \oplus B$$

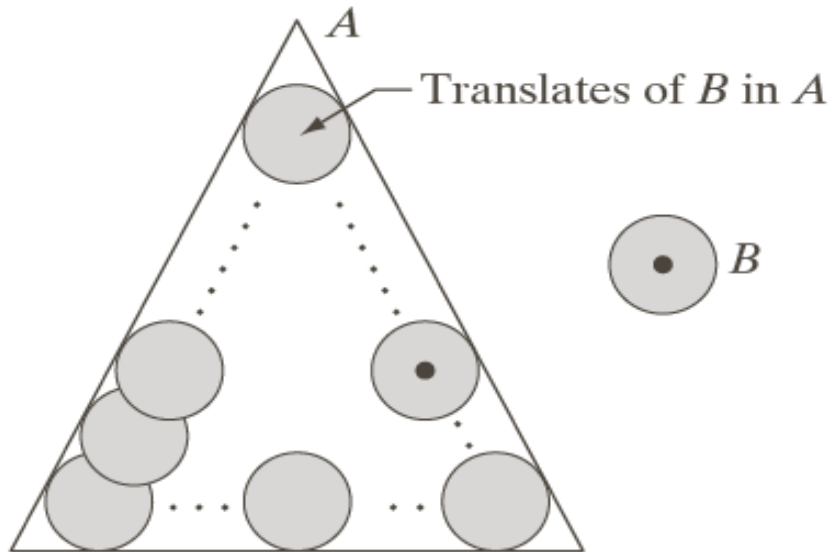


Opening is useful for:

- eliminates protrusions (حذف بیرون زدگی)
- breaks necks
- smoothes contour

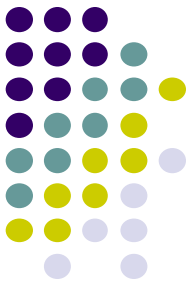


Example of Opening



a b c d

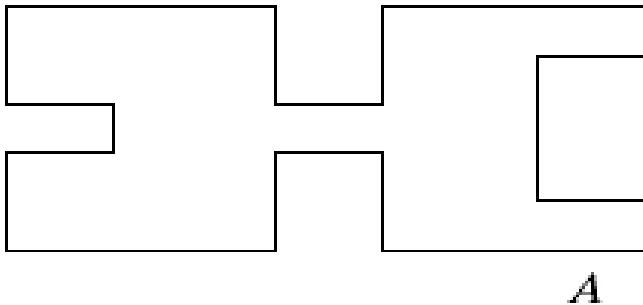
FIGURE 9.8 (a) Structuring element B “rolling” along the inner boundary of A (the dot indicates the origin of B). (b) Structuring element. (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded). We did not shade A in (a) for clarity.



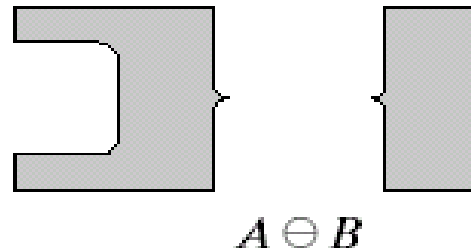
Another Example of Opening

The opening of image f by structuring element s , denoted $f \circ s$ is simply an erosion followed by a dilation

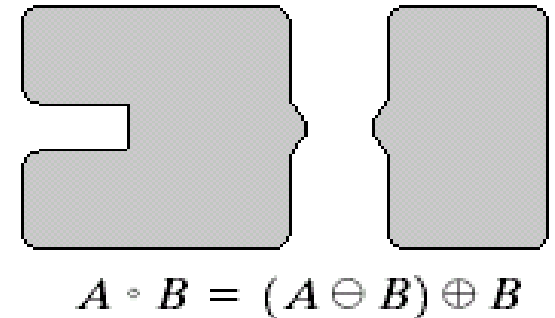
$$f \circ s = (f \ominus s) \oplus s$$



Original shape



After erosion

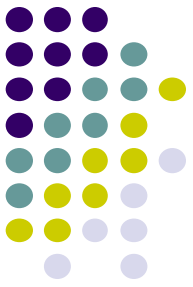


After dilation
(opening)

Note a disc shaped structuring element is used



Yet another Opening Example



Original
Image

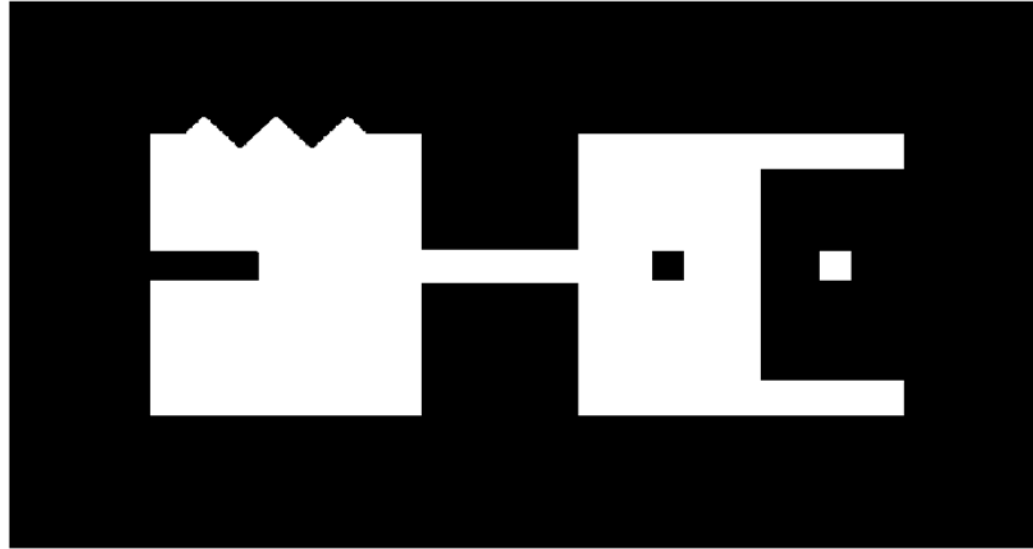
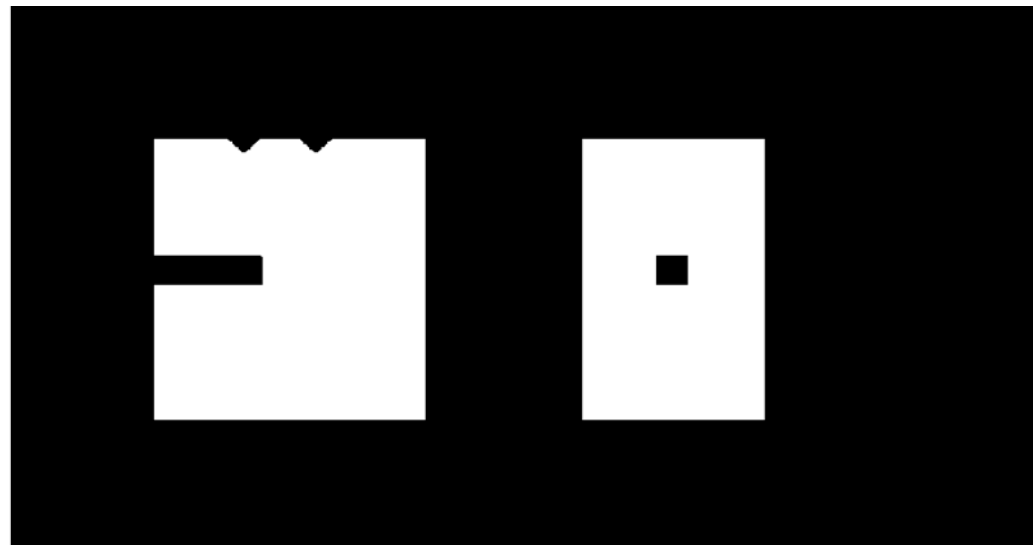
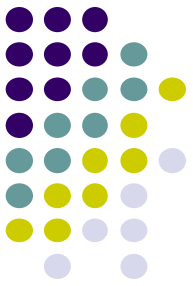


Image
After
Opening

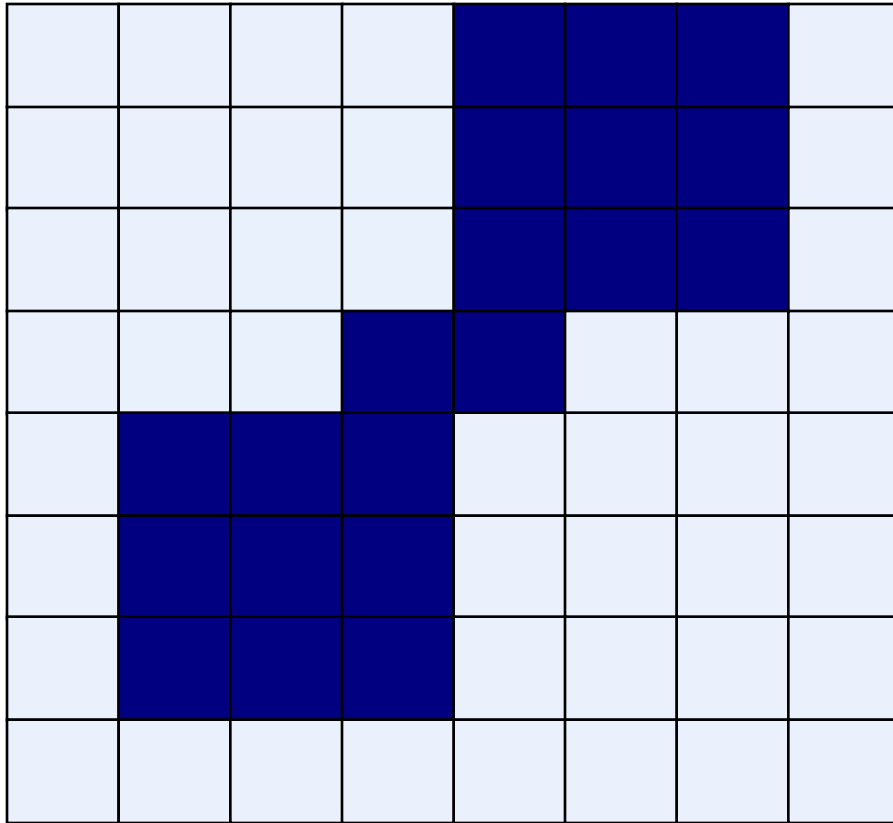




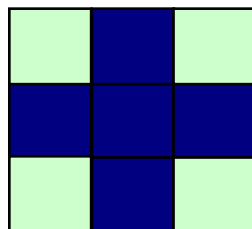
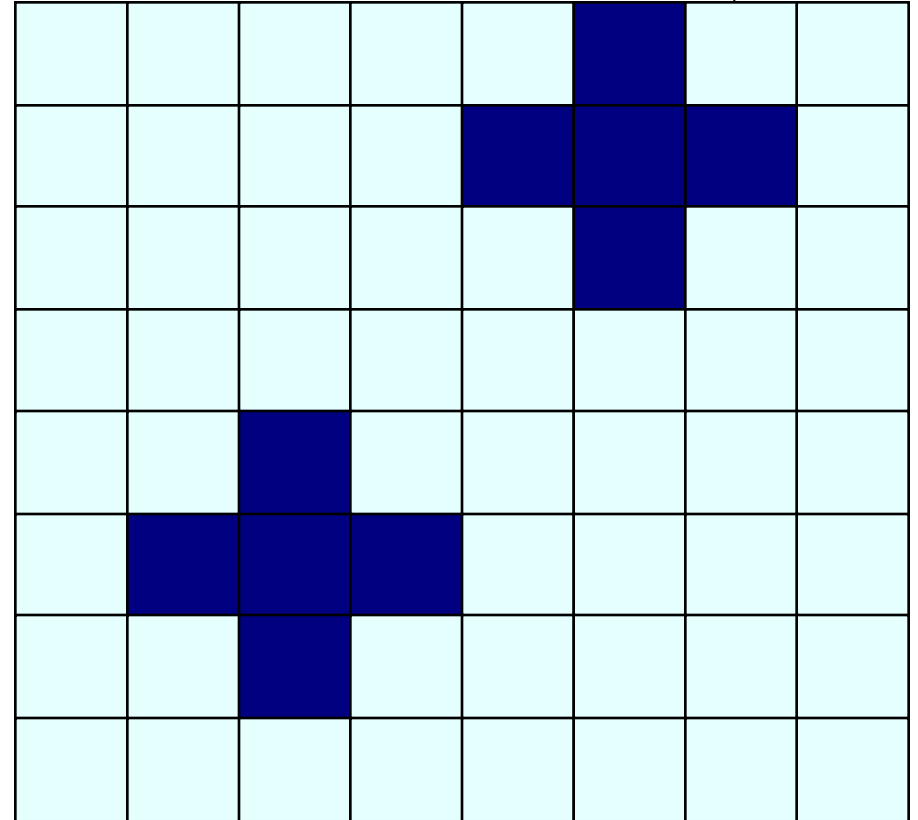
Example of Opening With Non-disc SE



Original Image



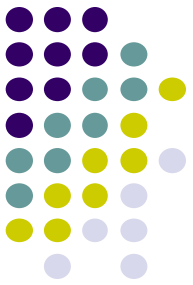
Processed Image



Structuring Element



A matlab example code for opening



% Remove the smaller objects in an image.

close all;

clear all;

% Read the image into the MATLAB workspace and display it.

% I = imread('blobs.png');

I = imread('snowflakes.png');

imshow(I)

% Create a disk-shaped structuring element with a radius of 5 pixels.

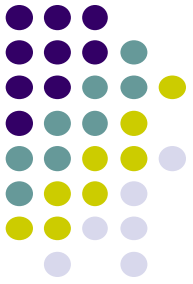
se = strel('disk',5);

% Remove flakes having a radius less than 5 pixels by opening it with

% the disk-shaped structuring element created in step 2.

I_opened = imopen(I,se);

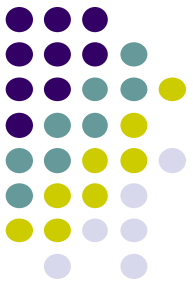
figure, imshow(I_opened,[])



Closing

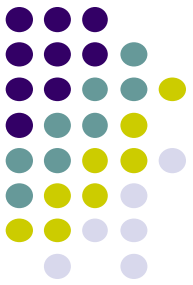
Is dilation followed by erosion, denoted •

$$A \bullet B = (A \oplus B) \ominus B$$

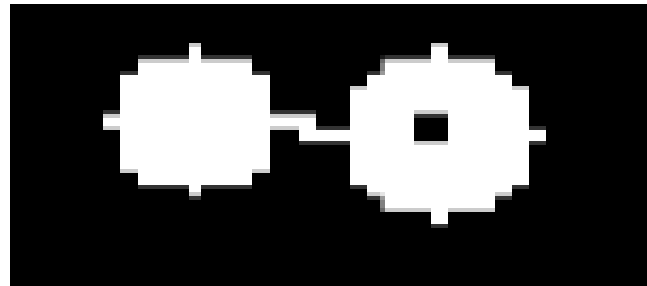


Closing Usefulness:

- smooth contour
- fuse narrow breaks and long thin gulfs
- eliminate small holes
- fill gaps in the contour



Opening & Closing



A



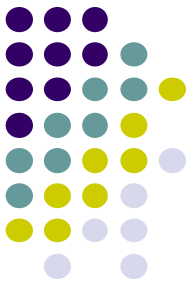
opening of A

→ removal of small protrusions, thin connections, ...

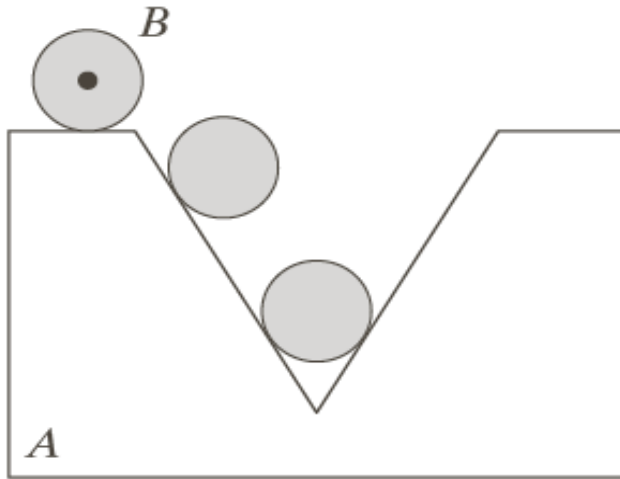


closing of A

→ removal of holes

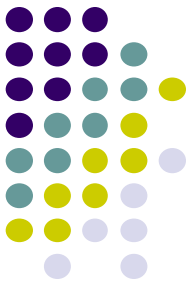


Example of Closing



a b c

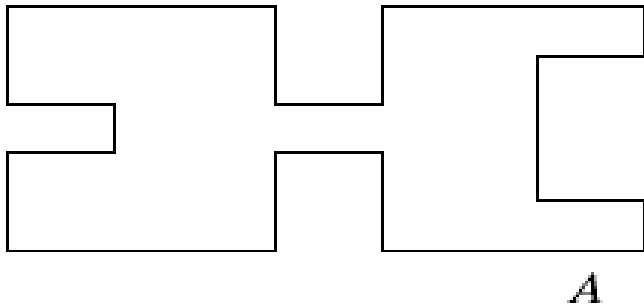
FIGURE 9.9 (a) Structuring element B “rolling” on the outer boundary of set A . (b) The heavy line is the outer boundary of the closing. (c) Complete closing (shaded). We did not shade A in (a) for clarity.



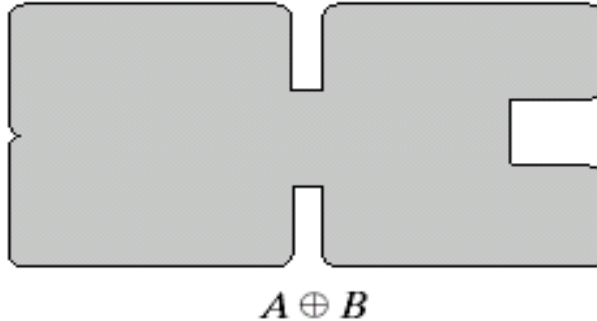
Another Example of Closing

The closing of image f by structuring element s , denoted $f \bullet s$ is simply a dilation followed by an erosion

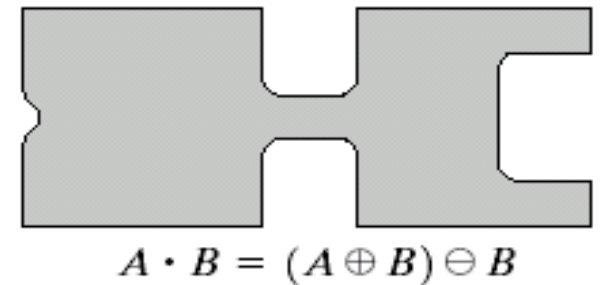
$$f \bullet s = (f \oplus s) \ominus s$$



Original shape



After dilation

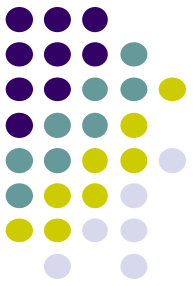


After erosion
(closing)

Note a disc shaped structuring element is used



Yet Another Closing Example



Original
Image

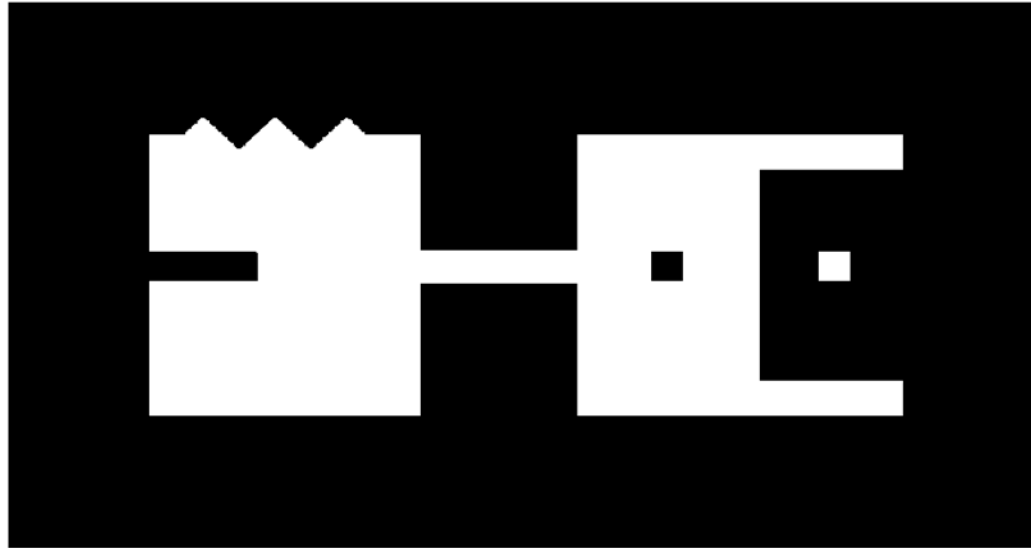
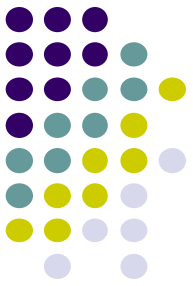


Image
After
Closing

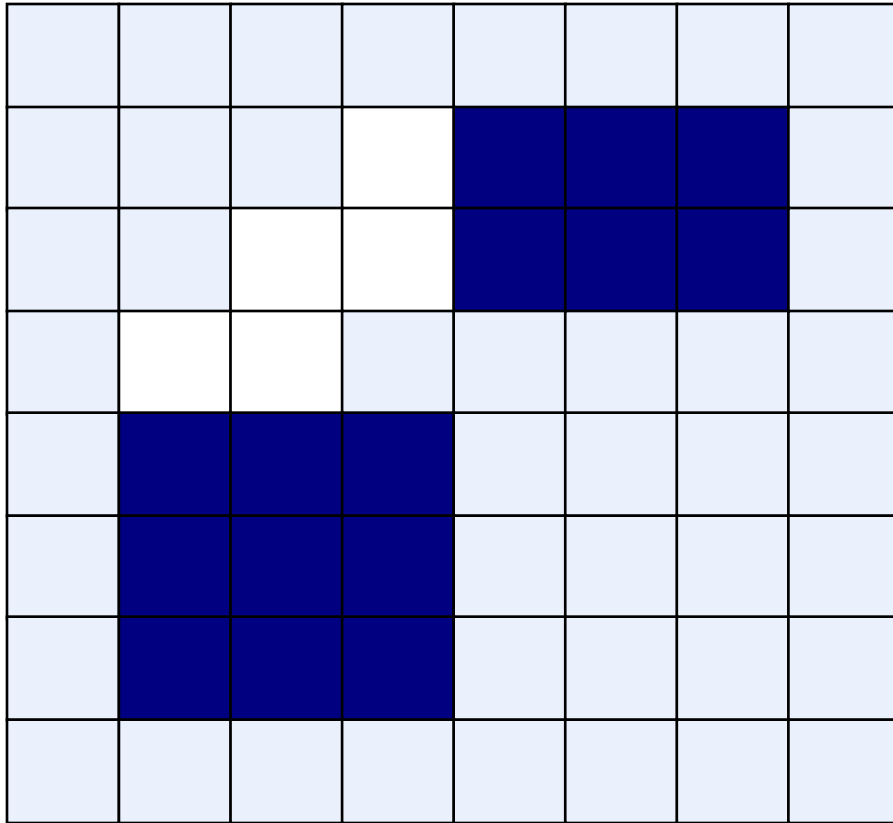




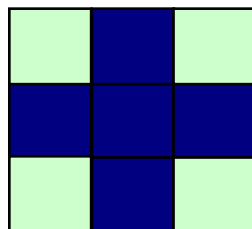
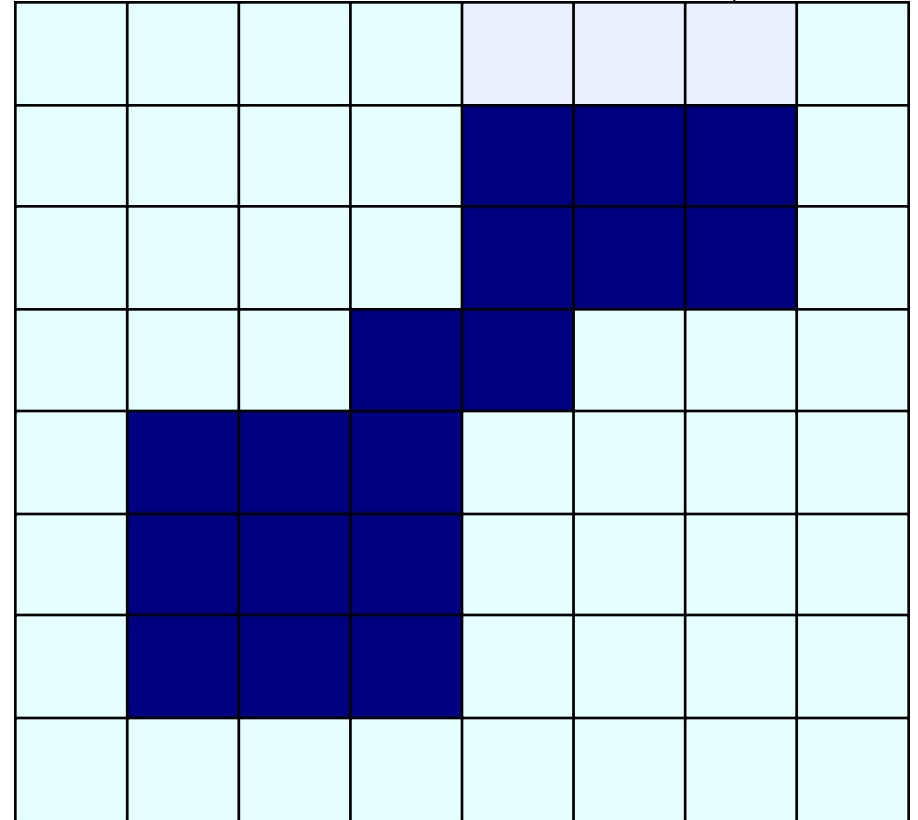
Example of Closing With Non-disc SE



Original Image



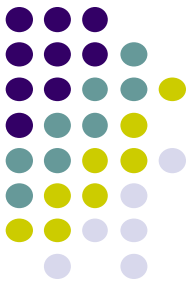
Processed Image



Structuring Element



A matlab example code for opening



**% Use `imclose` to join the circles in the image together by filling in the
% gaps between them and by smoothing their outer edges.**

close all;

clear all;

%Read the image into the MATLAB workspace and view it.

originalBW = imread('circles.png');

imshow(originalBW);

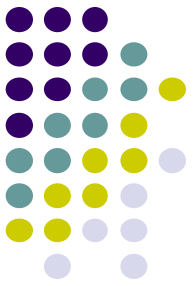
**% Create a disk-shaped structuring element. Use a disk structuring element
% to preserve the circular nature of the object. Specify a radius of 10
% pixels so that the largest gap gets filled.**

se = `strel('disk',10);`

% Perform a morphological close operation on the image.

closeBW = `imclose(originalBW,se);`

figure, imshow(closeBW)



Opening and Closing Properties

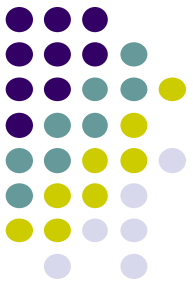
Opening

- (i) $A^\circ B$ is a subset (subimage) of A
- (ii) If C is a subset of D , then $C^\circ B$ is a subset of $D^\circ B$
- (iii) $(A^\circ B)^\circ B = A^\circ B$

Closing

- (i) A is a subset (subimage) of $A \bullet B$
- (ii) If C is a subset of D , then $C \bullet B$ is a subset of $D \bullet B$
- (iii) $(A \bullet B) \bullet B = A \bullet B$

Note: repeated openings/closings has no effect!

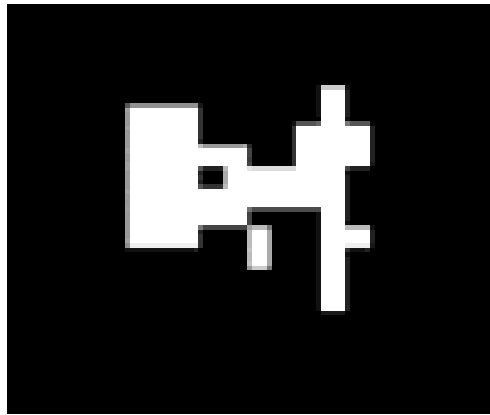


Duality

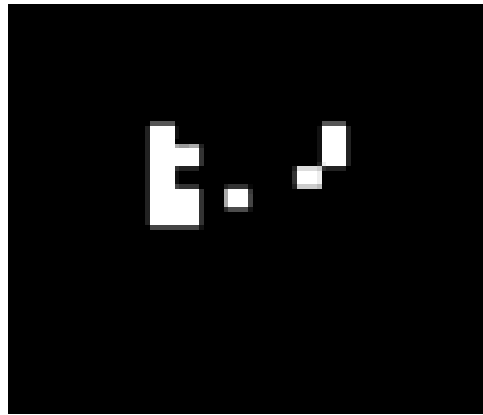
Between Opening and Closing

- Opening and closing are dual with respect to complementation and reflection

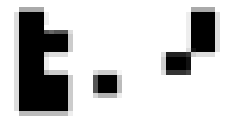
$$(A \bullet B)^c = (A^c \circ \hat{B})$$



A

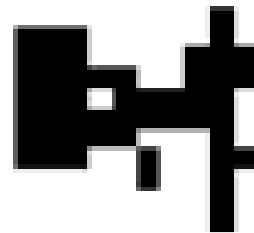
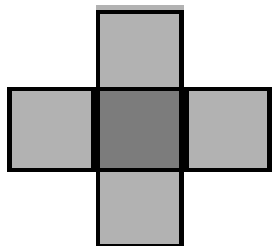


$A \ominus B$



$(A \ominus B)^C$

$$B = \hat{B}$$



A^C



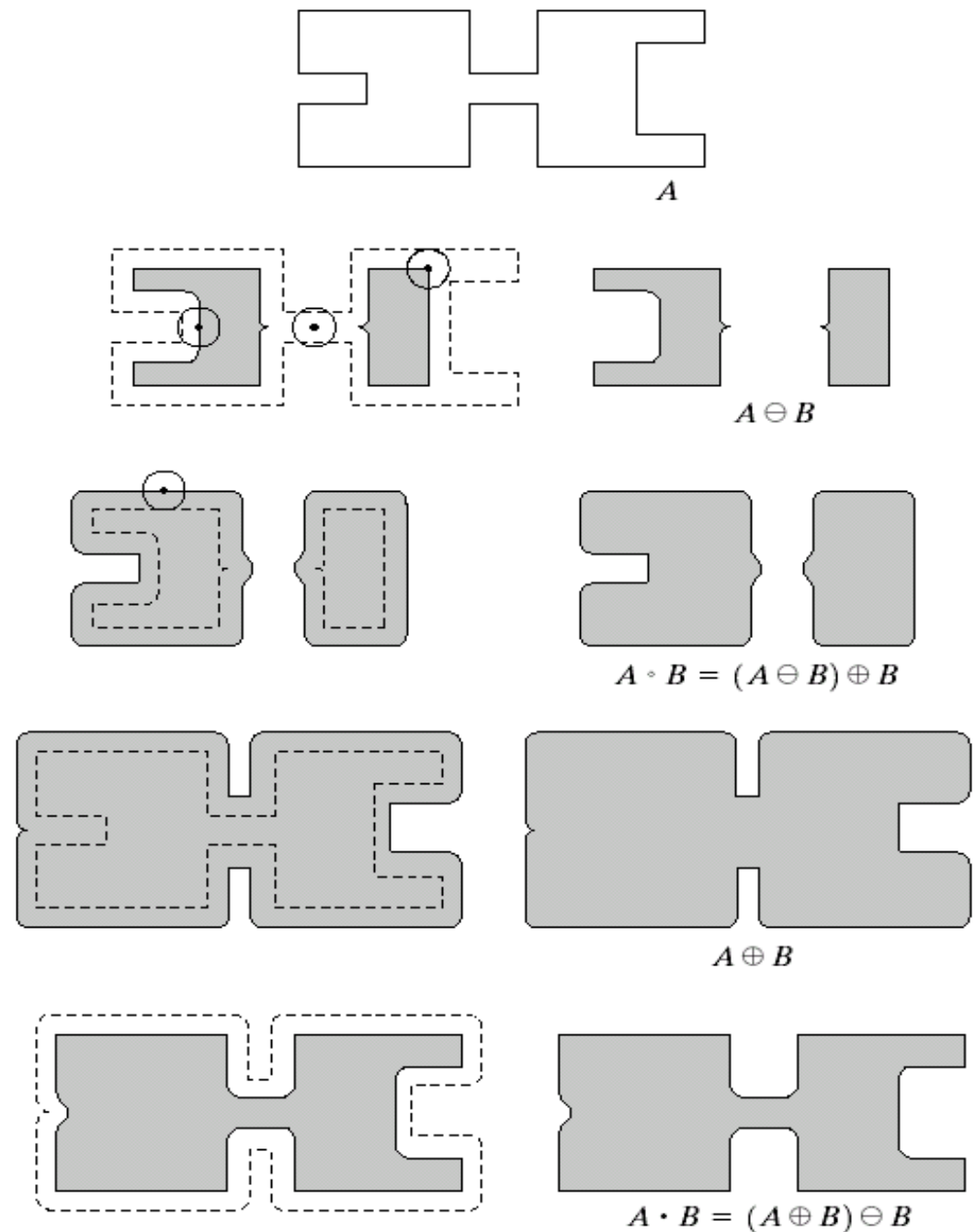
$A^C \oplus B$



a
b c
d e
f g
h i

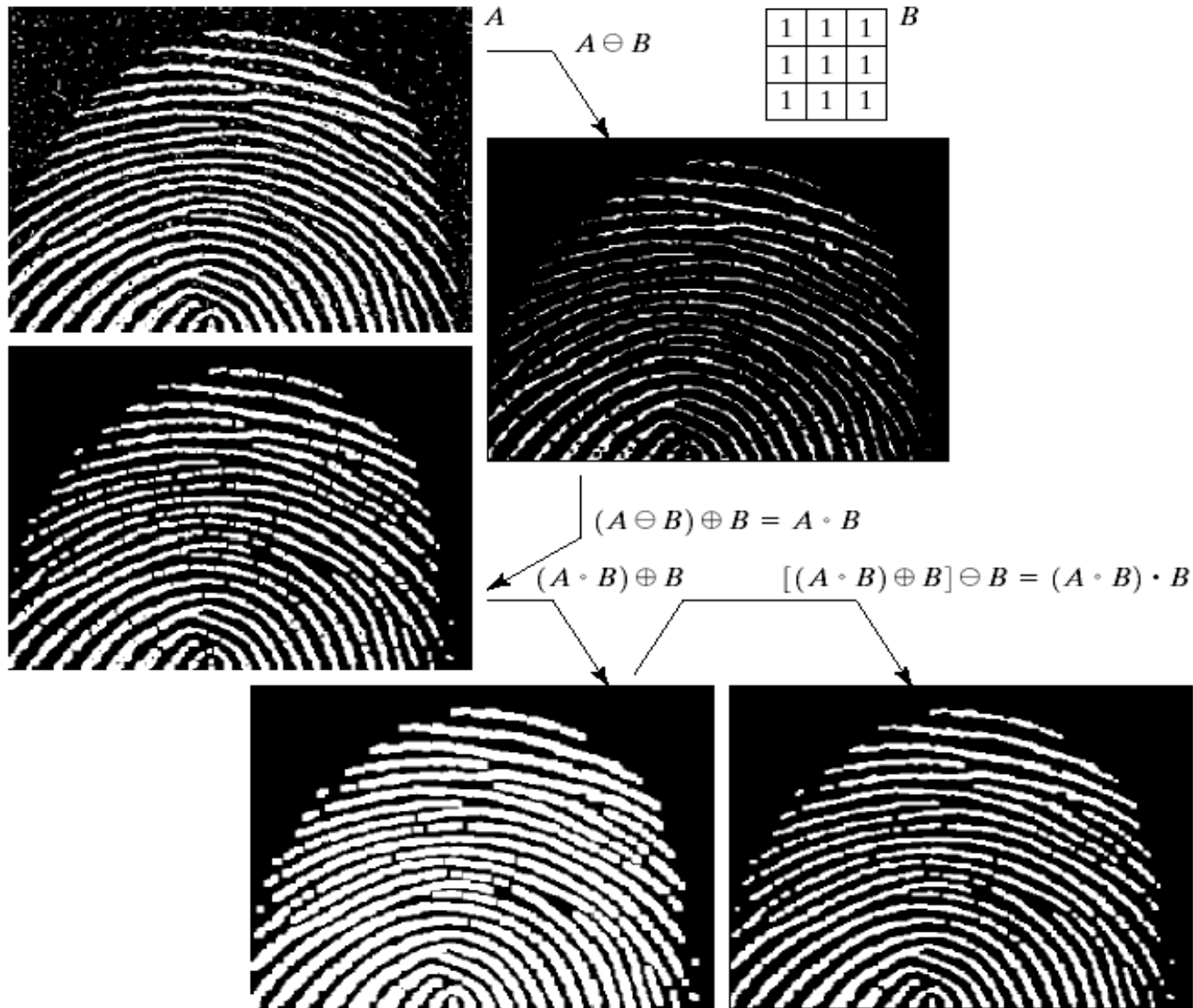
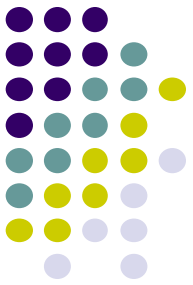
FIGURE 9.10
Morphological opening and closing. The structuring element is the small circle shown in various positions in (b). The dark dot is the center of the structuring element.

Text
Example of
Opening
and Closing



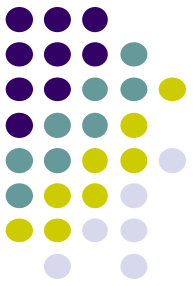


Opening and Closing at Work: Example

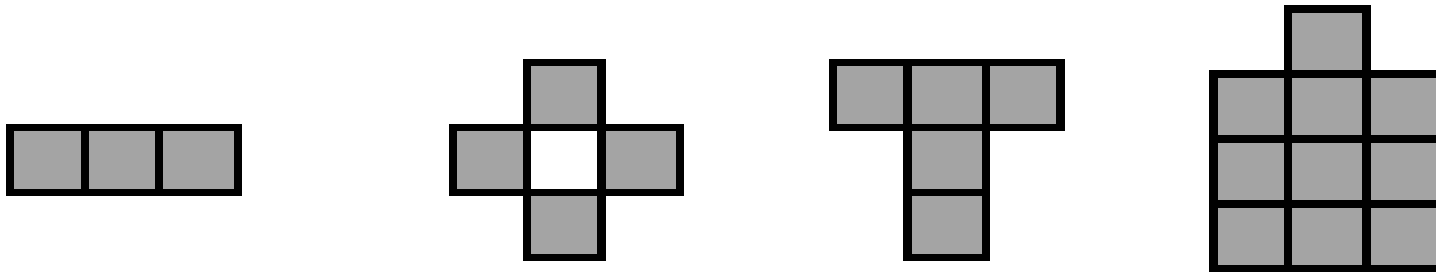




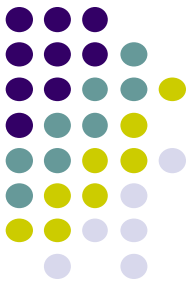
Hit-or-Miss Transformation $\odot$ (HMT)



- find location of one shape among a set of shapes
"template matching"



- composite SE: object part (B1) and background part (B2)
- does B1 *fits the object while, simultaneously*, B2 misses the object, i.e., *fits the background*?



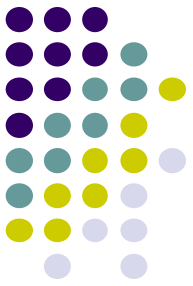
Hit-or-Miss Transform – More:

The hit-or-miss transform is a general binary morphological operation that can be used to look for particular patterns of foreground and background pixels in an image.

The hit-and-miss transform is a basic tool for shape detection.

Concept: To detect a shape:

- **Hit object**
- **Miss background**

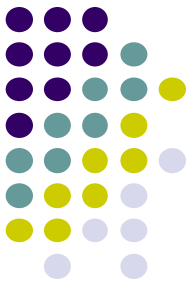


Hit-or-Miss Transform – More ...

- The structural elements used for Hit-or-miss transforms are an extension to the ones used with dilation, erosion etc.
- The structural elements can contain both foreground and background pixels, rather than just foreground pixels, i.e. both ones and zeros.
- The structuring element is superimposed over each pixel in the input image, and if an exact match is found between the foreground and background pixels in the structuring element and the image, the input pixel lying below the origin of the structuring element is set to the foreground pixel value. If it does not match, the input pixel is replaced by the boundary pixel value.



The hit-or-miss transform is defined as:



Let $B = \{B_1, B_2\}$, where B_1 is the set formed from elements of B associated with an object and B_2 is the set of elements of B associated with the corresponding background, where B_1 and B_2 are disjoint. Then HMT, or $A \circledast B$, is:

$$B = (B_1, B_2)$$

B_1 : object

B_2 : background

$$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$$



The Hit-or-Miss Transformation by picture:

if B denotes the set composed of D and its background, the match (or set of matches) of B in A , denoted $A \circledast B$,

$$A \circledast B = (A \ominus D) \cap [A^c \ominus (W - D)]$$

$$B = (B_1, B_2)$$

B_1 : object

B_2 : background

$$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$$

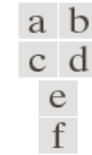
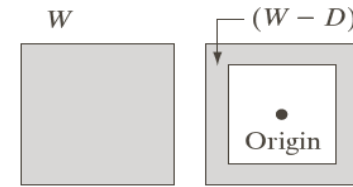
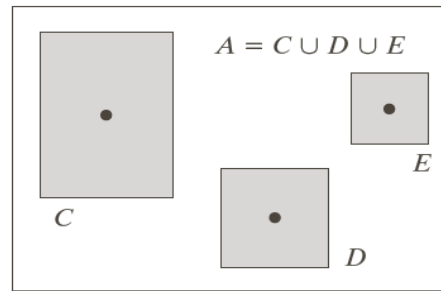


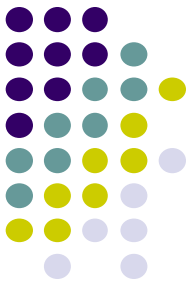
FIGURE 9.12

(a) Set A . (b) A window, W , and the local background of D with respect to W , $(W - D)$. (c) Complement of A . (d) Erosion of A by D . (e) Erosion of A^c by $(W - D)$. (f) Intersection of (d) and (e), showing the location of the origin of D , as desired. The dots indicate the origins of C , D , and E .

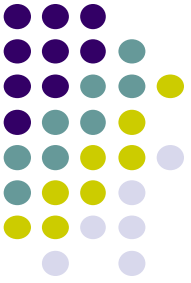


Binary Hit and Miss Transform

Example Code in Matlab:



```
bw = [0 0 0 0 0 0  
      0 0 1 1 0 0  
      0 1 1 1 1 0  
      0 1 1 1 1 0  
      0 0 1 1 0 0  
      0 0 1 0 0 0];  
interval = [0 -1 -1  
            1 1 -1  
            0 1 0];  
bw2 = bwHitMiss(bw,interval)
```



Thank you

for your attention